

Project Proposal

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Decision: Mental Health × COVID-19 infection rates

We have **committed to the Mental Health dataset** and dropped the US Supply Chain dataset (our initial analysis found essentially no signal in it and its headline risk variable was data leakage. see *Considered and rejected* at the end). We pair the Household Pulse mental-health data with **state-level COVID-19 case data** to ask whether pandemic intensity tracked the population's unmet need for mental-health support.

Our two focused research questions and their planned outputs are in [Research questions](#); the assignable analysis plan is in [Analysis plan](#). The core analysis is already implemented and producing results in [Covid_MentalHealth_analysis.qmd](#).

Mental Health Care x COVID-19 infection rates

<https://catalog.data.gov/dataset/mental-health-care-in-the-last-4-weeks>

1. Dataset & research questions

Source and licence

Mental-health outcome (primary dataset). The data comes from the **U.S. Census Bureau's Household Pulse Survey**, published by the **National Center for Health Statistics (NCHS / CDC)** and distributed through [data.gov](#). As a work of the US federal government it is in the **public domain** (U.S. Government Work / no copyright), so it is free to use, redistribute and analyse with attribution to the source.

COVID-19 exposure (second dataset). State-level COVID-19 case counts come from the **New York Times COVID-19 data** repository (<https://github.com/nytimes/covid->

[19-data](#)), file `us-states.csv` (saved locally as `Data/covid_us_states_nyt.csv`). It reports **cumulative** confirmed cases and deaths per US state per day. The NYT releases this data for **free use with attribution** for non-commercial purposes, which covers this academic project. We difference the cumulative counts into daily new cases and normalise by **2020 Census state population** to get new cases per 100,000.

What is in it

Each row reports **one estimate**: the percentage of US adults who, in the last 4 weeks, took a given mental-health-care action, for a particular subgroup, geography and time period.

Key variables:

- **Indicator** — one of four actions:
 - “Received Counseling or Therapy, Last 4 Weeks”
 - “Needed Counseling or Therapy But Did Not Get It, Last 4 Weeks”
 - “Took Prescription Medication for Mental Health, Last 4 Weeks”
 - “Took Prescription Medication for Mental Health And/Or Received Counseling or Therapy, Last 4 Weeks”
- **Group / Subgroup** — how the population is split (By Sex, By Age, National Estimate, By Presence of Symptoms of Anxiety/Depression, By Race/Hispanic ethnicity, By Education, By State, By Sexual orientation, By Disability status, By Gender identity).
- **State** — geography of the estimate.
- **Time Period** — the survey window (number, label, start and end dates).
- **Value, LowCI, HighCI** — the estimated percentage and its 95% confidence interval.
- **Suppression Flag** — marks estimates withheld for being statistically unreliable.

Research questions

We focus on **two** questions, both centred on whether COVID-19 infection intensity is linked to the population’s **need for mental-health support**.

For the results shown here we use the indicator **“Needed Counseling or Therapy But Did Not Get It, Last 4 Weeks”** as a concrete measure of unmet need, and we do find a signal there (below). We are **not yet committing to this single indicator**, though. The two questions below are stated for the current indicator but carry over to those variants unchanged.

1. **Exploratory / associational (primary)** — *Across US states and survey windows, is a higher COVID-19 case rate associated with a higher unmet need for mental-health support?*

- **Response variable:** `unmet_need` (% adults, By State). **Predictor:** `covid_rate` (new COVID cases per 100k during each survey window).
- **Method & why:** ordinary linear regression `lm(unmet_need ~ covid_rate)` — both variables are continuous and we want the direction/size of a linear association.
- **Null hypothesis:** no association, i.e. the slope on `covid_rate` is 0.
- **What a meaningful result looks like:** a slope reliably different from 0.
- **Planned output:** a **scatter plot** (“Unmet mental-health need vs COVID case rate, by state and survey window”) with a fitted line, plus the **slope coefficient**, **its 95% CI and p-value**. (*Implemented — current result: positive and significant but weak: slope $5.4e-4$, 95% CI $[3.4e-4, 7.5e-4]$, $p = 3e-7$, $R^2 = 0.02$.)*)
- **Inferential status:** this is **exploratory and ecological** (state-level aggregates, observational) — it generates a hypothesis about a link, it does **not** establish individual-level causation.

2. Descriptive (temporal) — How did national unmet need and the national COVID case rate move together over Aug 2020 – Apr 2022?

- **Variables:** national `unmet_need` and national `covid_rate` by survey window start date.
- **Method & why:** time-series description.
- **What a meaningful result looks like:** visible alignment (or clear non-alignment) between the two series across the pandemic waves.
- **Planned output:** a **two-panel time-series plot** of national unmet need and national COVID case rate on a shared time axis. (*Implemented.*)

Context and audience

The analysis targets a public-health and policy-oriented audience interested in how pandemic intensity related to unmet mental-health need across US states. Suitable presentation formats: a Quarto web report / project website and a chart-driven data story.

2. Initial data analysis

The **core joined analysis** (mental health x COVID) is implemented in [Covid_MentalHealth_analysis.qmd](#); the exploratory IDA of the mental-health dataset alone is in [Mental_Health_last_4_weeks.qmd](#). Our exploration covers:

- `dim()` and `unique()` on the structural columns (indicators, groups, number of time periods and geographies).
- Parsing **Time Period Start Date** into a real **Date** and confirming the date range.
- Counting suppressed/missing **Value** entries (the dataset’s main missingness mechanism).
- **Plot (trend):** national value of each indicator over the survey window.

- **Plot (relationship/between-group):** “Received counseling or therapy” by sex, and “Took prescription medication” by age, over time.

Data-quality observations: values are pre-aggregated percentages with confidence intervals; some estimates are suppressed for reliability (giving NA values); subgroup definitions are well documented; time coverage is limited to roughly Aug 2020 – Dec 2021.

3. Analysis plan

- Robustness: add a time lag and state fixed effects to RQ1; report whether the association strengthens, Short results note + updated coefficient
- Write-up & narrative: turn results into the report/website data story; caption every figure, Final Quarto report / site

Cleaning / transformation already handled: date parsing of survey windows; differencing cumulative COVID counts and flooring occasional negative corrections at 0; per-100k normalisation via 2020 Census populations; dropping suppressed estimates.

Main uncertainties / risks:

- **Ecological design.** Both datasets are state-level aggregates, so any association is between *states*, not individuals — we report associations, not individual causation.
- **Weak effect size.** RQ1 is significant but explains little variance ($R^2 = 0.02$); the lag / fixed-effects tests whether a cleaner relationship emerges.
- **Suppressed estimates** create non-random missingness for smaller states.
- **Reporting artefacts** in COVID counts (weekend dips, backfills) are smoothed by aggregating to the ~2-week survey window.

Considered and rejected — US Supply Chain Risk Analysis

We also explored the Kaggle *US Supply Chain Risk Analysis Dataset* (CC0; 1,000 records, 24 variables) for a delay-prediction project — full IDA in [US Supply Chain Risk Analysis.qmd](#). We **rejected it** because our initial modelling found essentially no signal: regressing `Delay_Days` on the available predictors gave $R^2 = 0.007$, adjusted R^2 negative, **F-test p = 0.63**, and no significant predictor; the only variable that “explained” delay, `Supply_Risk_Flag`, turned out to be **data leakage** (it is exactly 1 whenever `Delay_Days > 0`). The interesting predictive story simply does not exist in the data.

4. Workflow organisation

- **Code style.** We follow the tidyverse style guide. We'll use `styler` to format R code and agree as a group to write in a consistent style.
- **Packages & reproducibility.** `tidyverse` for cleaning and plots and `lubridate` for dates; a few extra packages may be added for tables or model output. We plan to use `renv` to pin package versions so the project runs on different machines.
- **Git & task division.** One shared GitHub repository. Each member works on their own or task-specific branch if needed, else (based on the small group size) we'll be working on different files but on the same branch.
- **.gitignore.** We exclude files that need not be shared: `.Rhistory`, `.RData`, temporary files, secrets such as `.env`, and large output files.

Next steps

Scope is now locked: the Mental Health $\times$ COVID analysis is implemented and producing results (RQ1 and RQ2 above). The remaining weeks go to robustness, presentation and the write-up.

Question for feedback: RQ1's association may be weak ($R^2 = 0.02$, with "Needed Counseling or Therapy But Did Not Get It, Last 4 Weeks"). Is a same-window state-level comparison the right framing, or should we prioritise the **lag / fixed-effects** refinement?